

progress!

Question F: The point that no one wants

Input file: standard input Output file: standard output

Time limit: 1 Sec Memory limit: 128 MB

Submission: 649 Resolution: 301

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Description of the topic

On a coordinate axis, from 1 to m In the interval, draw n Line segment, the starting and ending points of each line segment happen to be integers. Now we only consider integer points. So if an integer point does not fall between the starting point and the end point of any line segment (including the starting point and the end point itself), this integer point is not wanted by anyone.

Please calculate which integer points are not needed according to the given line segment.

Input

The first behavior is two integers n and m ;

Next, there is n Site, two integers per line x and y Indicates that the starting point of a line segment is x , the end point is y .

Output

The first behavior is an integer, indicating how many points no one wants in total;

The second behavior is that all the points that no one wants are output in order from small to large, separated by a space between each two numbers.

Sample input and output

Sample input #1

[Copy](#)

5 14 2 4 1 3 7 7 9 12 10 11

Sample output #1

[Copy](#)

5 5 6 8 13 14

Sample input #2

[Copy](#)

1 7 1 7

Sample output #2

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0

Data range

$$1 \leq n, m \leq 5000$$

$$1 \leq x \leq y \leq m$$

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